

Causality Detection via Symplectic Quandles

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September 2026

Abstract

We study whether symplectic quandle colorings can reveal causal structure encoded by “sky links”—links formed by the spheres of all light rays through two events in the space of light rays of a $(2 + 1)$ -dimensional globally hyperbolic spacetime. The testbed is the connected sum of two Hopf links H (causally unrelated events) and the infinite family of Allen–Swenberg links L_n , which the Alexander–Conway polynomial cannot tell apart from H . We use the enhanced quandle counting polynomial, in which every coloring is weighted by the size of the subquandle generated by its colors. For the 16-element symplectic quandle $T = (\mathbb{Z}/4)^2$ we prove that $|\text{Hom}(Q(L_n), T)| = 640 + 96 \cdot 16^n$ for every $n \geq 1$, while $|\text{Hom}(Q(H), T)| = 736$, and we determine the enhanced polynomial of every L_n . Hence already the plain quandle counting invariant distinguishes H from every Allen–Swenberg link, and the Allen–Swenberg links from each other. Analogous closed formulas hold for $(\mathbb{Z}/2)^4$ and for an 8-element degenerate symplectic quandle $(\mathbb{Z}/2)^3$ ($176 + 48 \cdot 16^n$ versus 224), and the quandle $(\mathbb{Z}_7)^2$ also separates H from the whole family. Over $(\mathbb{Z}_p)^2$ the enhanced polynomial carries at most one number beyond the counting invariant, and whether it separates L_1 from H depends on p : it does for $p = 7, 11, 17, 19, 23, 31$ and does not for $p = 2, 3, 5, 13$. The proofs combine an exact transfer-matrix description of the family with certified finite computations.

Keywords: quandle colorings; symplectic quandles; enhanced counting polynomial; globally hyperbolic spacetimes; causality; Allen–Swenberg links.

Mathematics Subject Classification (2020): Primary 57K12; Secondary 57K10, 83C75.

1 Introduction

Knots, in mathematical terms, are smooth embeddings of the circle S^1 into three-dimensional Euclidean space $\mathbb{R}^3$ (or into S^3). Two knots are considered equivalent if they are ambient isotopic; by Reidemeister’s theorem this happens exactly when their diagrams are related by a finite sequence of Reidemeister moves and planar isotopies. The most fundamental knot is the unknot (or trivial knot), which is simply a geometrically unknotted loop – a plain circle in space. When several circles are involved, we refer to the configuration as a link; a knot is a link with one component. Among links, the trivial link (or unlink) consists of two or more disjoint, unlinked loops. The simplest nontrivial link is the Hopf link, formed by two unknotted circles that cross each other twice in a diagram.

Knot invariants are functions that assign the same value to equivalent knots or links, although distinct knots may share the same value. Such invariants have emerged as tools for understanding causal structure in globally hyperbolic spacetimes. The Low conjecture [15] asserts that two events in a $(2 + 1)$ -dimensional globally hyperbolic spacetime whose Cauchy surface has universal cover

$\mathbb{R}^2$ are causally related if and only if their skies are linked in the space of light rays. Natário and Tod [17] formulated the Legendrian Low conjecture in dimension $3 + 1$: for a Cauchy surface homeomorphic to $\mathbb{R}^3$, two events are causally related if and only if their skies are Legendrian linked.

Both conjectures were proved by Chernov and Nemirovski [8]; the $(2 + 1)$ -dimensional statement holds whenever the Cauchy surface Σ is not homeomorphic to S^2 or $\mathbb{R}P^2$. Here and throughout, two events are *causally related* if there is a causal curve from one to the other, that is, a piecewise smooth curve whose tangent vectors are timelike or null; informally, information can travel between them without exceeding the speed of light. These results show a deep connection between the topology of sky links and the causal structure of spacetime, and motivate the use of link invariants as detectors of causality. Natário and Tod [17] examined specific pairs of causally related events and observed that the associated sky links have non-trivial Kauffman polynomial, but their results were limited to a particular class of skies.

Chernov, Martin and Petkova [7] observed that Khovanov homology (a categorification of the Jones polynomial) and Heegaard Floer homology (a categorification of the Alexander–Conway polynomial) detect causality in $(2 + 1)$ -dimensional globally hyperbolic spacetimes whose Cauchy surface is not S^2 or $\mathbb{R}P^2$.

The polynomial invariants themselves are weaker than their categorifications. Allen and Swenberg [1] conjectured that the Jones polynomial always detects causality in dimension $2 + 1$, and they showed that the Alexander–Conway polynomial cannot distinguish 2-sky-like links from H : they constructed an infinite family $\{L_n\}_{n \geq 1}$ of “2-sky-like” links whose Conway polynomial equals that of the connected sum of two Hopf links H —the sky link of causally unrelated events—while their Jones polynomials differ from that of H . Recently Chernov, Harper and Kohli [6] proved that the Links–Gould polynomial distinguishes every L_n from H . This limitation of the Alexander–Conway polynomial underscores the need for other computable invariants that detect causal relationships.

A quandle is an algebraic structure whose axioms mirror the Reidemeister moves, which makes it a natural source of knot and link invariants. Joyce [13] and, independently, Matveev [16] associated to a link L its fundamental quandle $Q(L)$; for a knot K the fundamental quandle determines K up to replacing it by its reversed mirror image. A practical way of producing computable invariants is to count quandle homomorphisms from $Q(L)$ to a fixed finite target quandle T , i.e. to compute $|\text{Hom}(Q(L), T)|$.

Nelson and collaborators refined this count by the enhanced quandle counting polynomial [20, 18], which records the cardinality of the image subquandle $\text{Im}(f) \subseteq T$ of each homomorphism $f \in \text{Hom}(Q(L), T)$. Symplectic quandles [18, 22] often contain rich, connected subquandles, which makes them natural candidates for this enhancement.

Recent research has explored whether quandle invariants can detect causality in cases where the Alexander–Conway polynomial fails. Leventhal [14] showed that counting invariants for affine Alexander quandles are insufficient; the Takasaki quandle is the special case $t = -1$ of an Alexander quandle. Jain [12] considered the symplectic quandle $(\mathbb{Z}_5)^2$. The first version of the present paper considered $(\mathbb{Z}_3)^2$ and repeated the $(\mathbb{Z}_5)^2$ computations of [12].

What goes wrong, and what is true. Both [12] and version 1 of this paper evaluated the enhanced polynomial by counting the number of distinct colors on the arcs of a diagram. Since $Q(L)$ is generated by the arcs, the image of a homomorphism f is the subquandle $\langle f(x_1), \dots, f(x_n) \rangle$ *generated* by the arc colors, which is in general larger than the set of arc colors; only the former is a link invariant. We make the following contributions.

1. We give an explicit Reidemeister II move on H that changes the “number of arc colors” polynomial (Example 5.1).

2. With the corrected definition, $\Phi_E(L_1, T) = \Phi_E(H, T)$ for $T = (\mathbb{Z}_3)^2$ and $T = (\mathbb{Z}_5)^2$ (Table 1); the separation claimed in version 1 and in [12] for L_1 disappears.
3. For $T = (\mathbb{Z}/4)^2$ with the standard symplectic form we prove, for all $n \geq 1$,

$$|\mathrm{Hom}(Q(L_n), T)| = 640 + 96 \cdot 16^n, \quad |\mathrm{Hom}(Q(H), T)| = 736,$$

and we compute $\Phi_E(L_n, T)$ for all n (Theorem 5.6). Hence T distinguishes H and all L_n pairwise. For $(\mathbb{Z}/2)^4$ we obtain $736 + 480 \cdot 100^n$ (Theorem 5.7), and for the 8-element degenerate symplectic quandle $(\mathbb{Z}/2)^3$ with form $J \oplus 0$ we obtain $176 + 48 \cdot 16^n$ versus 224 for H (Theorem 5.8).

4. We prove a corrected “transfer” principle: for every coloring quandle admitting identity colorings of the Allen–Swenberg block, $|\mathrm{Hom}(Q(L_n), T)|$ is non-decreasing in n (Lemma 5.4). As a consequence $(\mathbb{Z}_7)^2$ separates H from every L_n , and $(\mathbb{Z}_5)^2$ separates H from every L_n with $n \geq 2$.
5. We verify that the link data are correct: from the Wirtinger presentations we reconstruct planar diagrams of L_1, L_2, L_3 whose Kauffman-bracket spans are 116, 216, 316, as predicted by Allen and Swenberg (Section 5.2).

The structure of the paper is as follows. Section 2 reviews causal structure in globally hyperbolic spacetimes. Section 3 recalls quandles and the corrected enhanced counting polynomial. Section 4 introduces symplectic quandles and proves the structural facts we need. Section 5 contains the computations and the main results, and Section 6 concludes.

2 Spacetime Causality

A spacetime X is a time-oriented Lorentzian manifold. In suitable local coordinates at a point the metric is the Lorentz product

$$(x_1, \dots, x_n, t_1) \cdot (y_1, \dots, y_n, t_2) = x_1 y_1 + \dots + x_n y_n - t_1 t_2.$$

A nonzero tangent vector v is timelike if $v \cdot v < 0$, null if $v \cdot v = 0$ and causal if $v \cdot v \leq 0$. Two points (events) $x, y \in X$ are causally related if there is a piecewise smooth curve γ from x to y whose tangent vectors are causal and all future-directed (or all past-directed). Such a curve is called causal; it describes a trajectory that does not exceed the speed of light.

A Cauchy surface Σ is a subset of the spacetime such that every inextendible causal curve intersects Σ exactly once. Informally, Σ carries complete information about the whole evolution of the spacetime. A spacetime is globally hyperbolic if it has a Cauchy surface, see [11]. These are the most studied spacetimes, since one version of the Strong Cosmic Censorship conjecture of Penrose [21] asserts that all physically reasonable spacetimes are globally hyperbolic.

Theorem 2.1 (Geroch [10], Bernal and Sánchez [2, 3]). *A globally hyperbolic spacetime X is homeomorphic to $\Sigma \times \mathbb{R}$, where every slice $\Sigma \times \{t\}$ is a Cauchy surface. Moreover X is diffeomorphic to $\Sigma \times \mathbb{R}$ with the $\mathbb{R}$ -coordinate a smooth time function whose level sets are smooth spacelike Cauchy surfaces.*

Geroch [10] established the topological splitting, Bernal and Sánchez [2] showed that the Cauchy surface can be taken smooth and spacelike, and in [3] they obtained the smooth splitting by a temporal function whose level sets are smooth spacelike Cauchy surfaces.

Theorem 2.2 (Bernal and Sánchez [4]). *A spacetime X is globally hyperbolic if and only if*

- *it is causal, i.e. it contains no closed causal curves (no time travel), and*
- *for all $x, y \in X$ the set $J^+(x) \cap J^-(y)$ is compact, where $J^+(x)$ and $J^-(y)$ denote the causal future of x and the causal past of y .*

The second condition is sometimes described informally as the absence of naked singularities.

The space $\mathcal{N}$ of future-directed light rays (unparametrized null geodesics) of a globally hyperbolic spacetime is identified with the spherical cotangent bundle $ST^*\Sigma$ of a Cauchy surface; when $\Sigma = \mathbb{R}^2$ this is the solid torus $S^1 \times \mathbb{R}^2$. The light rays through an event x form a sphere $S_x \subset \mathcal{N}$, the *sky* of x . The sky is determined by where the light emitted at x is seen at a given moment, i.e. on a level set of a time function. Low’s conjecture relates causality to the linking of skies in $\mathcal{N}$.

Theorem 2.3 (Low conjecture; proved by Chernov and Nemirovski [8]). *Let (X, g) be a $(2 + 1)$ -dimensional globally hyperbolic spacetime whose Cauchy surface is not homeomorphic to S^2 or $\mathbb{R}P^2$ (equivalently, its universal cover is $\mathbb{R}^2$). Then two events $x, y \in X$ are causally related if and only if their skies $S_x \cup S_y$ form a non-trivial link in $\mathcal{N}$.*

The implication “causally unrelated $\Rightarrow$ skies unlinked” is the easier half and was known before [8]. In other words, causality corresponds to linking in the solid torus: x and y are causally related if and only if $S_x \cup S_y$ is not isotopic to the sky link of two causally unrelated events.

Theorem 2.4 (Legendrian Low conjecture [17]; proved in [8]). *Two events x, y in a $(3 + 1)$ -dimensional globally hyperbolic spacetime with Cauchy surface homeomorphic to $\mathbb{R}^3$ are causally related if and only if their skies are Legendrian linked.*

3 Quandle Invariants of Links and Knots

Definition 3.1. A quandle is a non-empty set X with a binary operation $\triangleright$ satisfying

1. $x \triangleright x = x$ for all $x \in X$;
2. for every $y \in X$ the map $x \mapsto x \triangleright y$ is a bijection of X ; that is, for all $y, z \in X$ there is a unique $x \in X$ with $x \triangleright y = z$, which we denote $x = z \triangleright^{-1} y$;
3. $(x \triangleright y) \triangleright z = (x \triangleright z) \triangleright (y \triangleright z)$ for all $x, y, z \in X$.

The first axiom is idempotency: every element acts trivially on itself. The second guarantees that right multiplication can be undone: $(x \triangleright y) \triangleright^{-1} y = x$. The third axiom is right self-distributivity. Unlike groups, quandles are typically non-commutative and non-associative, so the order and grouping of operations matter. The three axioms correspond to the three Reidemeister moves, which makes quandles effective algebraic tools for distinguishing knots and links (see [19]).

Joyce [13] formalized the connection between knots and quandles by assigning quandle elements to arcs, with the operation encoding how arcs interact at each crossing, as illustrated in Figure 1.

In Figure 1, the left diagram shows a negative crossing, where the arc x passes under y from left to right, yielding $x \triangleright^{-1} y$. The right diagram shows a positive crossing, with x passing under y from right to left, yielding $x \triangleright y$. In both cases the arc on the right of the over-arc (with respect to its orientation) is acted on by $\triangleright y$ to give the arc on the left; so every crossing can be written as a single relation $a \triangleright b = c$, where b is the over-arc and a is the under-arc on its right. This is the form in which all relations in this paper are stored.

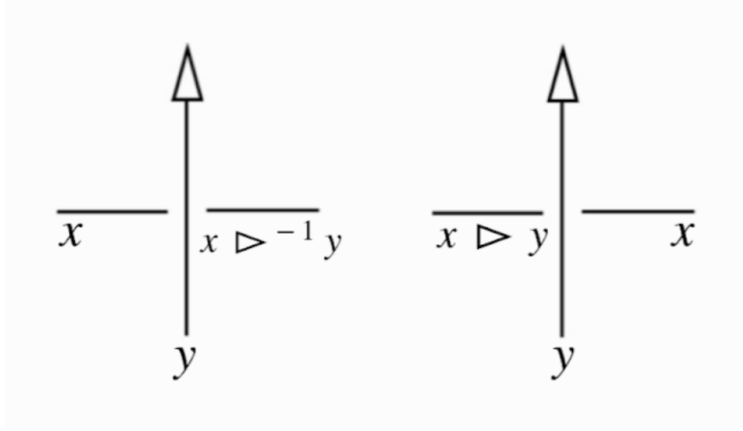


Figure 1: The quandle crossing relations.

Definition 3.2. Let L be an oriented link diagram with arc set A . The fundamental quandle $Q(L)$ is the quandle generated by A subject to one relation $a \triangleright b = c$ at every crossing.

Up to isomorphism $Q(L)$ does not depend on the diagram.

Definition 3.3. Let X_1 and X_2 be quandles. A map $\varphi: X_1 \rightarrow X_2$ is a quandle homomorphism if $\varphi(x \triangleright y) = \varphi(x) \triangleright \varphi(y)$ for all $x, y \in X_1$. A quandle isomorphism is a bijective homomorphism.

Theorem 3.4 (Joyce [13], Matveev [16]). *For knots $K, K' \subset S^3$ the fundamental quandles $Q(K)$ and $Q(K')$ are isomorphic if and only if there is a homeomorphism of S^3 , possibly reversing orientation, taking K to K' ; equivalently, K' is equivalent to K or to the reversed mirror image of K .*

Since the fundamental quandle is hard to compare directly, one uses homomorphisms into a finite target quandle T . Assigning to each arc of L an element of T and imposing the crossing relations gives a system of equations whose solutions form $\text{Hom}(Q(L), T)$.

Definition 3.5. A quandle coloring of L by T is an assignment of elements of T to the arcs of a diagram of L that satisfies $a \triangleright b = c$ at every crossing. Colorings are in bijection with $\text{Hom}(Q(L), T)$, and

$$\phi_T(L) = |\text{Hom}(Q(L), T)|$$

is a link invariant, the quandle counting invariant.

The counting invariant ignores structural information about each homomorphism. Quandle 2-cocycle invariants [5] weight each homomorphism by a cocycle. Nelson [20] introduced a two-variable subquandle polynomial, and Navas and Nelson [18] the enhanced quandle counting invariant, which records the size of the image of every homomorphism.

Definition 3.6. The enhanced quandle counting polynomial of L with respect to a finite quandle T is

$$\Phi_E(L, T) = \sum_{f \in \text{Hom}(Q(L), T)} q^{|\text{Im}(f)|}.$$

Here $\text{Im}(f) = f(Q(L))$ is the image of the whole fundamental quandle. It is a subquandle of T , and since $Q(L)$ is generated by the arcs $x_1, \dots, x_n$ of any diagram,

$$\text{Im}(f) = \langle f(x_1), \dots, f(x_n) \rangle,$$

the subquandle of T generated by the arc colors. For a finite quandle, the subquandle generated by a set S is the smallest subset containing S that is closed under $\triangleright$ (closure under $\triangleright^{-1}$ is then automatic, since right multiplications are permutations of finite order).

Remark 3.7. The set of colors $\{f(x_1), \dots, f(x_n)\}$ that actually appear on the arcs of a particular diagram is in general a proper subset of $\text{Im}(f)$, and its size depends on the diagram. Indeed, a Reidemeister II move creates a new arc whose color is $a \triangleright b$ for two existing colors a, b , and this element need not have appeared before. Hence the polynomial $\sum_f q^{\#\{f(x_1), \dots, f(x_n)\}}$ is *not* a link invariant. Version 1 of this paper, and [12], computed this diagram-dependent quantity. An explicit example is given in Example 5.1. (We thank Zining Fan and V. Chernov for pointing this out.)

Common choices of coloring quandles include:

Takasaki kei: $\mathbb{Z}_n$ with $x \triangleright y = 2y - x \pmod{n}$.

Affine Alexander quandle: a module over $\mathbb{Z}[t, t^{-1}]$ with $x \triangleright y = tx + (1 - t)y$. For $t = -1$ it reduces to the Takasaki quandle.

Conjugation quandle: a group G with $x \triangleright y = y^{-1}xy$.

In this paper we use symplectic quandles, introduced in the next section.

4 Symplectic Quandle

A symplectic quandle is a quandle structure on a module defined by an antisymmetric bilinear form. It was introduced as a generalization of earlier constructions (see [18, 22]).

Definition 4.1. Let R be a finite commutative ring, $M = R^r$, and let $\langle \cdot, \cdot \rangle: M \times M \rightarrow R$ be a bilinear form with $\langle x, x \rangle = 0$ for all $x \in M$ (so it is antisymmetric). We write $\langle x, y \rangle = xJy^T$ for an $r \times r$ matrix J with $J^T = -J$ and zero diagonal. Then M is a quandle with

$$x \triangleright y = x + \langle x, y \rangle y, \quad x \triangleright^{-1} y = x - \langle x, y \rangle y.$$

Indeed $x \triangleright x = x$, and $(x \triangleright y) \triangleright^{-1} y = x + \langle x, y \rangle y - \langle x + \langle x, y \rangle y, y \rangle y = x$ because $\langle y, y \rangle = 0$; self-distributivity follows from the fact that each $R_y: x \mapsto x \triangleright y$ is a symplectic transvection, so $R_y(x \triangleright z) = R_y(x) \triangleright R_y(z)$. Unless stated otherwise, J is the standard form $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ (block-diagonal for $r > 2$). We write $\mathbb{Z}_n = \mathbb{Z}/n$, and $(\mathbb{Z}_n)^2$ (also $(\mathbb{Z}/n)^2$) for the rank-2 symplectic quandle over $\mathbb{Z}_n$ with the standard form; $(\mathbb{Z}/2)^4$ carries the form $J \oplus J$. Degenerate forms are allowed: the quandle

$$T_8 = (\mathbb{Z}/2)^3 \quad \text{with} \quad \langle x, y \rangle = x_1y_2 + x_2y_1 \quad (\text{the form } J \oplus 0)$$

has 8 elements and will play a role in Section 5.7.

Definition 4.2. A quandle is connected if any two elements are related by a finite sequence of operations $\triangleright y, \triangleright^{-1} y$.

Theorem 4.3 (Navas and Nelson [18]). *Let M be a symplectic quandle over a finite field with a non-degenerate form. Then the subquandle $M \setminus \{0\}$ is connected.*

The fundamental quandle of a knot is connected, while the fundamental quandle of a link with m components has exactly m orbits, one for each component. (Version 1 stated incorrectly that the fundamental quandle of every nontrivial link is connected.) Over a field a non-degenerate antisymmetric form requires even rank.

We now collect the facts about symplectic quandles that are used below.

Lemma 4.4 (Scaling). *Let R be a commutative ring and $\lambda \in R^\times$. The symplectic quandles (R^{2k}, J) and $(R^{2k}, \lambda J)$ are isomorphic.*

Proof. Let $A = \text{diag}(\lambda^{-1}I_k, I_k)$ in a symplectic basis $e_1, \dots, e_k, f_1, \dots, f_k$ with $\langle e_i, f_j \rangle = \delta_{ij}$. Then $\langle Ax, Ay \rangle = \lambda^{-1}\langle x, y \rangle$, so $Ax + \lambda\langle Ax, Ay \rangle Ay = A(x + \langle x, y \rangle y)$, i.e. $x \mapsto Ax$ is an isomorphism $(R^{2k}, J) \rightarrow (R^{2k}, \lambda J)$. $\square$

In particular, by Lemma 4.4, the quandle $M = (\mathbb{Z}_3)^2$ with $J' = 2J$ used in version 1 is isomorphic to $(\mathbb{Z}_3)^2$ with J ; scaling the form is not a genuine perturbation.

Lemma 4.5 (Automorphisms). *Every $g \in SL(2, R)$ acts on (R^2, J) by quandle automorphisms, and $f \mapsto g \circ f$ preserves $\text{Hom}(Q(L), T)$ and the size of $\text{Im}(f)$.*

Proof. $\langle gx, gy \rangle = \det(g)\langle x, y \rangle = \langle x, y \rangle$, hence $g(x \triangleright y) = gx \triangleright gy$. $\square$

Lemma 4.6 (The zero color). *In any symplectic quandle, $0 \triangleright y = 0$ and $x \triangleright 0 = x$ for all x, y . Consequently, if one arc of a component K of L is colored 0 then every arc of K is colored 0, and colorings of L with $K \mapsto 0$ are in bijection with colorings of the sublink $L \setminus K$.*

Proof. The identities are immediate from the definition. Along K , consecutive arcs satisfy $x_{\text{out}} = x_{\text{in}} \triangleright^{\pm 1} y$, so $x_{\text{in}} = 0$ forces $x_{\text{out}} = 0$ and conversely. At crossings where K is the over-arc the relation becomes $a \triangleright 0 = a$, which simply identifies the two under-arcs, exactly as in a diagram of $L \setminus K$. $\square$

Lemma 4.7 (Subquandles of $(\mathbb{F}_p)^2$). *Let p be a prime and $T = (\mathbb{F}_p)^2$. If $S \subseteq T$ contains two linearly independent vectors, then the subquandle generated by S contains $T \setminus \{0\}$. Consequently every subquandle of T is either contained in a line $\mathbb{F}_p v$ (and is then a trivial quandle), or equals $T \setminus \{0\}$ or T .*

Proof. Let $a, c \in S$ be independent, $d = \langle a, c \rangle \neq 0$. The generated subquandle is invariant under the group generated by R_a and R_c , since $R_{x \triangleright y} = R_y R_x R_y^{-1}$. In the basis (a, c) these are the elementary matrices $\begin{pmatrix} 1 & -d \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ d & 1 \end{pmatrix}$, whose powers give all elementary matrices over the prime field $\mathbb{F}_p$; they generate $SL(2, p)$, which acts transitively on $T \setminus \{0\}$. If all elements of S lie on a line, then $\langle x, y \rangle = 0$ for $x, y \in S$ and S is already closed. $\square$

In particular a subquandle of $(\mathbb{Z}_5)^2$ has at most 5 elements or exactly 24 or 25 elements. Image sizes such as 21 or 22, which appear in version 1 and in [12], cannot occur for $\text{Im}(f)$; they are numbers of distinct arc colors.

5 Causality Detection using Symplectic Quandles

5.1 Sky links, H , and the Allen–Swenberg family

When the Cauchy surface is $\mathbb{R}^2$, the space of light rays is the solid torus $V = S^1 \times \mathbb{R}^2$ and the sky of an event is a circle in V . For two causally unrelated events the sky link is isotopic to two parallel fibers $S^1 \times \{p\}$, $S^1 \times \{q\}$. To study links in V with tools for links in S^3 one identifies V with the complement of an unknot $\mu \subset S^3$ and adds μ as a third component (version 1 incorrectly said “deleting”). Each fiber links μ once and the two fibers are unlinked, so two causally unrelated skies give the connected sum of two Hopf links

$$H = K_1 \cup \mu \cup K_2,$$

a chain in which μ is the middle component (Figure 2(c)).

Allen and Swenberg [1] considered 3-component “2-sky-like” links $K_1 \cup K_2 \cup \mu$ with μ an unknotted meridian and $\text{lk}(K_i, \mu) = \pm 1$, and constructed the infinite family

$$L_n = C(1 * T_0(n)), \quad n \geq 1,$$

where T_0 is a tangle whose Conway fraction vanishes, i.e. $\nabla(N(T_0)) = 0$ and $\nabla(D(T_0)) = 1$ for its numerator and denominator closures, and $T_0(n)$ is the sum of n copies of T_0 (an explicit description of T_0 as a concatenation of two clasped tangles is given by Chernov, Harper and Kohli [6]). Allen and Swenberg consider two closures $C_{\pm}$ with the same invariants below; the checks in Section 5.2 confirm that the diagrams used here (Figures 2(a),(b)) have these invariants, and all our statements concern these diagrams. They proved $\nabla(L_n) = \nabla(H) = z^2$ for all n , while the span of the Kauffman bracket of L_n equals $100n + 16$ and that of H equals 16. Following their proposal we treat “causality detection” operationally: an invariant provides evidence for detecting causality on this testbed if it separates the Allen–Swenberg links from H . We work with the labelled diagrams of L_1 and L_2 in Figures 2(a),(b) and with the family L_n generated from them (Section 5.2).

5.2 Link diagrams, presentations, and their verification

For each oriented diagram we assign a generator x_i to each arc between undercrossings and impose one relation $a \triangleright b = c$ per crossing (Figure 1). For H (Figure 2(c)):

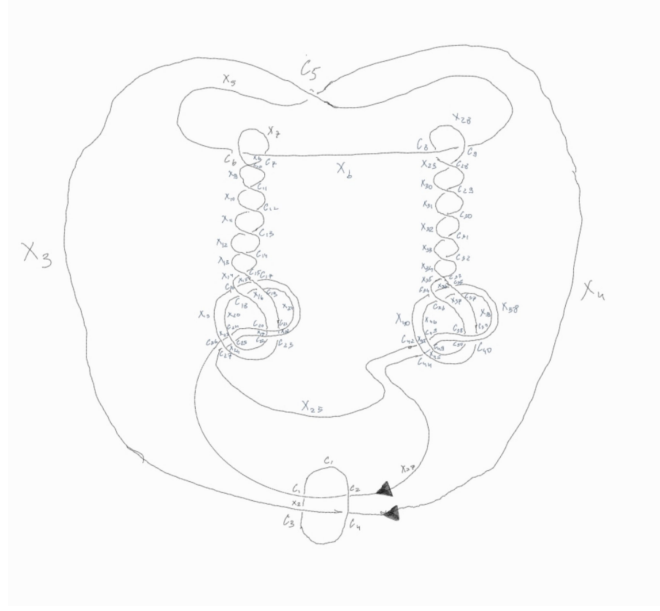
$$x_1 \triangleright x_3 = x_1, \quad x_3 \triangleright x_1 = x_2, \quad x_2 \triangleright x_4 = x_3, \quad x_4 \triangleright x_3 = x_4.$$

The presentations of L_1 (45 relations) and L_2 (85 relations) are those of the published notebooks of version 1 (<https://github.com/harryfrences-svg/Horizon-Research>), read off from Figures 2(a),(b); they are listed in Appendix A. Because the whole paper rests on these data, we checked them independently.

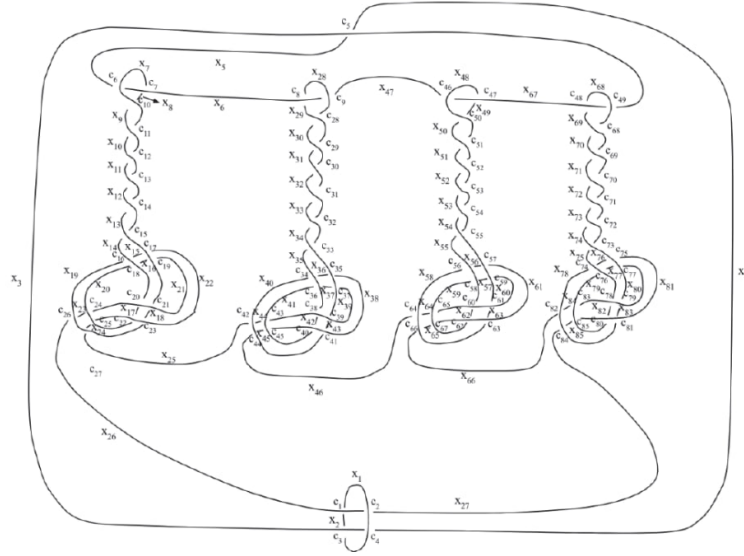
1. *Components and linking numbers.* Following under-arcs through crossings, L_1 has three components with 2 (the axis μ), 39 and 4 arcs; L_2 has 2, 77 and 6 arcs. Computing the linking numbers once from the crossings where K_i passes under K_j and once from those where K_j passes under K_i gives in both cases $\text{lk}(\mu, K_1) = \text{lk}(\mu, K_2) = 1$ and $\text{lk}(K_1, K_2) = 0$.
2. *Alexander polynomials.* By Fox calculus (with $m_c = m_b^{-1} m_a m_b$ for $a \triangleright b = c$), the one-variable Alexander matrix of each of H, L_1, L_2 has a first minor equal to $(t - 1)^2$ up to units, and the multivariable Alexander polynomial of each link equals $t_{\mu} - 1$ up to units. So the presentations have exactly the Alexander invariants required of the Allen–Swenberg links.
3. *Planar reconstruction and the Jones polynomial.* A Wirtinger presentation records the over-arc of every crossing and which under-arc lies to its right, but not the order in which an over-arc passes over its crossings. For L_1 we searched all $8 \times 589,824$ choices of component orientations and over-crossing orders and kept those whose 4-valent ribbon graph is planar ($V - E + F = 2$). Exactly one diagram (up to reversing all orientations) is planar, so the presentation comes from a unique classical diagram. A randomized search finds planar realizations for L_2 and for the next member L_3 of the family (125 crossings). Their Kauffman brackets, computed by a state-sum algorithm, have spans

$$116 = 100 \cdot 1 + 16, \quad 216 = 100 \cdot 2 + 16, \quad 316 = 100 \cdot 3 + 16,$$

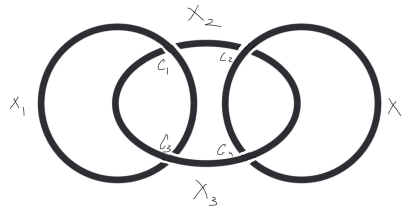
exactly as proved by Allen and Swenberg for L_1, L_2, L_3 . The planar diagrams are also accepted by SnapPy [9], which confirms that the sublinks $\mu \cup K_i$ are Hopf links.



(a) Labelled Allen–Swenberg link L_1 (45 crossings).



(b) Labelled Allen–Swenberg link L_2 (85 crossings).



(c) Labelled connected sum of two Hopf links H .

Figure 2: The test links. In (a) and (b) the axis μ is the small circle at the bottom (arcs x_1, x_2).

The family L_n . The diagram of L_2 is that of L_1 with a 40-crossing block B (crossings $c_{46}, \dots, c_{85}$, one copy of T_0) inserted between the outputs of the crossings c_9, c_{44} and the arcs x_3, x_{27} that run into the clasp with μ . The block is a 4-ended tangle with inputs x_{47}, x_{46} and outputs x_3, x_{27} , and all its over-arcs are internal. Inserting $n - 1$ copies of B produces a presentation of L_n ; for $n = 2$ this reproduces the presentation of L_2 verbatim, and for $n = 3$ the bracket span 316 confirms the construction.

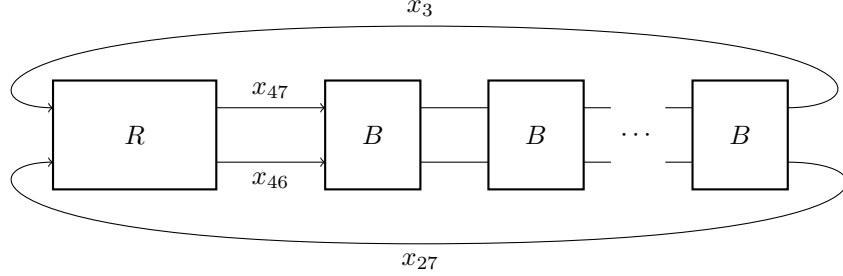


Figure 3: Schematic of L_n : the part R (crossings $c_1, \dots, c_{45}$ of L_2 , i.e. L_1 cut open at x_3, x_{27}) followed by $n - 1$ copies of the block B . The picture only records how the four boundary arcs are glued, not the geometry.

5.3 Computational method

All computations are exact and were done in Python 3 (package `symq`, see Reproducibility). For a coloring quandle T with operation table $\triangleright$ we enumerate $\text{Hom}(Q(L), T)$ by constraint propagation: each arc carries a set of admissible colors, each crossing $a \triangleright b = c$ is a constraint that removes a color from an arc only if no triple $(x, y, x \triangleright y)$ with $x, y, x \triangleright y$ admissible uses it, and the search branches on the arc with the fewest admissible colors. No coloring can be lost, and every complete assignment is checked against *all* crossing relations (residual test). A second implementation of the same algorithm in C is used for speed; for H, L_1, L_2 over $(\mathbb{Z}_3)^2$, $(\mathbb{Z}/4)^2$ and T_8 both implementations return identical sets of colorings. For each coloring we compute $|\text{Im}(f)|$ as the closure of the set of arc colors under $\triangleright$ (Definition 3.6).

As a check against version 1 we also computed the diagram-dependent arc-color count: we reproduce every number of version 1 (Table 1),¹ which shows that the difference between the two versions is due to the definition and not to the solver.

5.4 The error in version 1

Example 5.1 (Reidemeister II changes the arc-color count). Push the left circle of H under the right circle through the outer region. The arc x_1 is cut into x_1, x_6, x_5 and the relation $x_1 \triangleright x_3 = x_1$ is replaced by

$$x_5 \triangleright x_3 = x_1, \quad x_1 \triangleright x_4 = x_6, \quad x_5 \triangleright x_4 = x_6.$$

Every coloring extends uniquely ($x_6 = x_1 \triangleright x_4, x_5 = x_1$), so $|\text{Hom}|$ and $|\text{Im}(f)|$ are unchanged. Over $(\mathbb{Z}_3)^2$, however, the 48 colorings with $\mu \mapsto 0$ and linearly independent colors $a = f(x_1), c = f(x_4)$ use three colors on the original diagram and four colors on the new one, since $a \triangleright c = a + \langle a, c \rangle c$ is new. The arc-color polynomial changes from $9q + 72q^2 + 72q^3$ to $9q + 72q^2 + 24q^3 + 48q^4$, while the

¹Except the misprint $840q^3$ in the L_1 row of Table 2 of version 1, whose coefficients sum to $1705 \neq 1225$; the text of version 1 has the correct $360q^3$.

corrected polynomial is $9q + 72q^2 + 24q^3 + 48q^9$ in both cases. Over $(\mathbb{Z}_5)^2$ the arc-color polynomial changes from $25q + 360q^2 + 840q^3$ to $25q + 360q^2 + 360q^3 + 480q^4$.

For H the corrected polynomial can be written down for every prime.

Proposition 5.2. *Let p be a prime and $T = (\mathbb{Z}_p)^2$. Then $|\text{Hom}(Q(H), T)| = 2p^4 - p^2$ and*

$$\Phi_E(H, T) = p^2 q + 3p(p^2 - 1) q^2 + p(p - 2)(p^2 - 1) q^3 + p(p - 1)(p^2 - 1) q^{p^2}.$$

Proof. Over a field, $a \triangleright b = a$ if and only if $\langle a, b \rangle b = 0$, i.e. a and b are linearly dependent. The relations of H therefore force $x_2 = x_3 =: b$ and say exactly that $a = f(x_1)$ and $c = f(x_4)$ are dependent on b . If $b = 0$ there are p^4 choices of (a, c) ; if $b \neq 0$ then $a, c \in \mathbb{F}_p b$, giving $(p^2 - 1)p^2$ colorings. By Lemma 4.7, $\text{Im}(f) = \{a, b, c\}$ unless $b = 0$ and a, c are independent, in which case $\text{Im}(f) = T$ (it contains 0); there are $(p^2 - 1)(p^2 - p)$ such colorings. Counting the sizes of $\{a, b, c\}$ in the remaining cases gives the coefficients: $|\{a, b, c\}| = 1$ for p^2 colorings, $= 2$ for $3(p^2 - 1) + 3(p - 1)(p^2 - 1)$, and $= 3$ for $(p^2 - 1)(p - 2) + (p^2 - 1)(p - 1)(p - 2)$. $\square$

T	L	arc-color count (v1; not invariant)	corrected $\Phi_E(L, T)$
$(\mathbb{Z}_3)^2$ $ \text{Hom} = 153$	H	$9q + 72q^2 + 72q^3$	$9q + 72q^2 + 24q^3 + 48q^9$
	L_1	$9q + 72q^2 + 24q^3 + 48q^9$	$9q + 72q^2 + 24q^3 + 48q^9$
	L_2	$9q + 72q^2 + 24q^3 + 48q^9$	$9q + 72q^2 + 24q^3 + 48q^9$
$(\mathbb{Z}_5)^2$ $ \text{Hom} :$ 1225, 1225, 13225	H	$25q + 360q^2 + 840q^3$	$25q + 360q^2 + 360q^3 + 480q^{25}$
	L_1	$25q + 360q^2 + 360q^3 + 360q^{21} + 120q^{22}$	$25q + 360q^2 + 360q^3 + 480q^{25}$
	L_2	$\dots + 360q^{21} + 120q^{22} + 12000q^{25}$	$25q + 360q^2 + 360q^3 + 12480q^{25}$

Table 1: Version 1 versus the corrected invariant. The middle column reproduces version 1 (for L_2 over $(\mathbb{Z}_5)^2$, where version 1 reported no value, the first terms are as for L_1). With the correct definition L_1 and H have the same polynomial over $(\mathbb{Z}_3)^2$ and $(\mathbb{Z}_5)^2$.

Table 1 shows that the conclusion of version 1 does not survive: over $(\mathbb{Z}_3)^2$ nothing is separated, and over $(\mathbb{Z}_5)^2$ the corrected invariant separates L_2 from H and from L_1 (already the plain count does so) but not L_1 from H . The same applies to the $(\mathbb{Z}_5)^2$ computations of [12], which report exactly the arc-color polynomial of L_1 in the middle column. (For L_2 , [12] reports $|\text{Hom}| = 13125$ and a coefficient $1200q^{25}$; our computation gives $13225 = 25 + 360 + 360 + 360 + 120 + 12000$, so these appear to be misprints.)

The mechanism is transparent. By Lemma 4.6, the colorings with $\mu \mapsto 0$ are the colorings of the sky sublink $K_1 \cup K_2$. For H this sublink is the unlink and its colorings are the pairs $(a, c) \in T^2$. For L_1 over $(\mathbb{Z}_3)^2$ and $(\mathbb{Z}_5)^2$ the number of such colorings is also $|T|^2$; the only effect of the more complicated diagram is that the same colorings use many more distinct colors on the arcs, which version 1 mistook for a difference of the invariant.

5.5 Corrected results over $(\mathbb{Z}_p)^2$

By Lemma 4.7, over $(\mathbb{Z}_p)^2$ the corrected polynomial of any link can only differ from that of H in the coefficients of q^{p^2-1} and q^{p^2} , because colorings whose image lies in a line are colorings by a trivial quandle and their number depends only on the number of components. So the enhancement adds at most one number to the count, namely how the non-collinear colorings split between q^{p^2-1} and q^{p^2} . In every case where we enumerated them ($p \leq 13$ for L_1 , $p \leq 7$ for L_2), the colorings of L

with $\mu \mapsto v \neq 0$ are exactly those of H (all colors lie on the line $\mathbb{F}_p v$), so every additional coloring has $\mu \mapsto 0$ and uses the color 0. In these cases

$$\Phi_E(L, (\mathbb{Z}_p)^2) = \Phi_E(H, (\mathbb{Z}_p)^2) + (|\text{Hom}(Q(L), T)| - |\text{Hom}(Q(H), T)|) q^{p^2} \quad (L = L_1, L_2),$$

and the corrected enhancement adds nothing to the counting invariant. Note also that the unlink of two components has exactly $|T|^2$ colorings; since L_1 has more colorings with $\mu \mapsto 0$ over $(\mathbb{Z}_7)^2$, its sky sublink $K_1 \cup K_2$ is not the unlink. Table 2 lists the counts. We count the colorings with $\mu \mapsto 0$ using Lemma 4.5: fixing the arc x_7 of K_1 to be 0 or $e_1 = (1, 0)$, their number is $N_0 + (p^2 - 1)N_{e_1}$, where N_0 and N_{e_1} are the numbers of colorings with $\mu \mapsto 0$ and $x_7 \mapsto 0$, resp. $x_7 \mapsto e_1$ (both are enumerated; $N_0 = p^2$ in all cases).

p	H		L_1		L_2	
	$\mu \mapsto 0$	$ \text{Hom} $	$\mu \mapsto 0$	$ \text{Hom} $	$\mu \mapsto 0$	$ \text{Hom} $
2	16	28	16	28	16	28
3	81	153	81	153	81	153
5	625	1 225	625	1 225	12 625	13 225
7	2 401	4 753	120 673	123 025	7 464 289	7 466 641
11	14 641	29 161	458 161	472 681	75 413 041	$\geq 75 413 041$
13	28 561	56 953	28 561	56 953	–	–
17	83 521	166 753	12 147 265	$\geq 12 147 265$	–	–
19	130 321	260 281	11 074 321	$\geq 11 074 321$	–	–
23	279 841	559 153	84 219 169	$\geq 84 219 169$	–	–
29	707 281	1 413 721	707 281	–	–	–
31	923 521	1 846 081	58 062 721	$\geq 58 062 721$	–	–

Table 2: Counting invariants over $(\mathbb{Z}_p)^2$. The columns “ $\mu \mapsto 0$ ” count colorings with the axis colored 0, i.e. colorings of the sky sublink $K_1 \cup K_2$ (Lemma 4.6); for H this is p^4 . Where the colorings with $\mu \mapsto v \neq 0$ were not enumerated, only the lower bound $|\text{Hom}| \geq \#\{\mu \mapsto 0\}$ is given; it already exceeds $|\text{Hom}(Q(H), T)|$, except for $p = 29$, where the colorings with $\mu \mapsto 0$ number p^4 as for H and the rest was not enumerated. For $p \leq 13$ (L_1) and $p \leq 7$ (L_2) the colorings with $\mu \mapsto v \neq 0$ number exactly $(p^2 - 1)p^2$, as for H , and all their colors lie on the line $\mathbb{F}_p v$.

Theorem 5.3. *The counting invariant of the rank-2 symplectic quandle $(\mathbb{Z}_p)^2$ distinguishes L_1 from H for $p = 7, 11, 17, 19, 23, 31$ and does not for $p = 2, 3, 5, 13$. It distinguishes L_2 from H for $p = 5, 7, 11$ and does not for $p = 2, 3$. Over $(\mathbb{Z}_7)^2$ the corrected enhanced polynomials are*

$$\begin{aligned} \Phi_E(H) &= 49q + 1008q^2 + 1680q^3 + 2016q^{49}, \\ \Phi_E(L_1) &= 49q + 1008q^2 + 1680q^3 + 120288q^{49}, \\ \Phi_E(L_2) &= 49q + 1008q^2 + 1680q^3 + 7463904q^{49}. \end{aligned}$$

The set of primes that detect L_1 is not an interval, and it does not seem to contain all large primes: for $p = 29$ the colorings of L_1 with $\mu \mapsto 0$ number exactly 29^4 , as for H (the colorings with $\mu \mapsto v \neq 0$ were not enumerated for this p), while for $p = 31$ the colorings with $\mu \mapsto 0$ alone already outnumber all colorings of H . In all our data the additional colorings are colorings of the sky sublink $K_1 \cup K_2$, which is therefore not the unlink; see Conjecture 5.9 for a possible explanation of the exceptional primes.

5.6 Transfer along the Allen–Swenberg family

Version 1 claimed that every coloring of L_1 transfers to L_n “with the same image” and deduced that the enhanced polynomial separates H from the whole family. As stated, the argument uses the number of arc colors and a degree bound $\deg \Phi_E(H) = 3$, both of which fail for the correct invariant (e.g. $\deg \Phi_E(H, (\mathbb{Z}_5)^2) = 25 = \deg \Phi_E(L_1, (\mathbb{Z}_5)^2)$). There is also a gap in the construction: copying the colors of the first copy of T_0 into a new copy gives a coloring of the new block whose input colors are those *entering* the first copy, while inserting a block requires a coloring whose inputs and outputs are both equal to the colors *leaving* it. This is exactly the hypothesis of Lemma 5.4 below. The idea can be made precise as follows.

Cut L_2 into the part R (crossings $c_1, \dots, c_{45}$) and the block B (crossings $c_{46}, \dots, c_{85}$) along the four boundary arcs (Figure 3). For a finite quandle T and boundary colors $w = (w_1, w_2)$, $u = (u_1, u_2) \in T^2$ let

$$\begin{aligned} M_T[w, u] &= \#\{\text{colorings of } B \text{ with } (x_{47}, x_{46}) = w, (x_3, x_{27}) = u\}, \\ R_T[w, u] &= \#\{\text{colorings of } R \text{ with } (x_{47}, x_{46}) = w, (x_3, x_{27}) = u\}. \end{aligned}$$

Since the block has no over-arcs in common with the rest of the diagram, colorings of L_n are obtained by gluing colorings of R and of $n - 1$ copies of B along equal boundary colors, so

$$|\text{Hom}(Q(L_n), T)| = \sum_{w, u \in T^2} R_T[w, u] (M_T^{n-1})[w, u] \quad (n \geq 1). \quad (1)$$

For $n = 1, 2$ (and $n = 3$ over $(\mathbb{Z}/4)^2$) formula (1) agrees with the direct counts in every case we computed.

H is the member $n = 0$. The part R consists of the clasp C (crossings $c_1, \dots, c_5$, where μ meets the skies) followed by one copy of T_0 (crossings $c_6, \dots, c_{45}$). The presentation of this copy is isomorphic to that of the block B by a relabelling of arcs that sends its boundary arcs $x_5, x_{26}, x_{47}, x_{46}$ to $x_{47}, x_{46}, x_3, x_{27}$; we found the isomorphism by computer and checked it relation by relation. Let A_T be the transfer matrix of the clasp, from the colors of (x_3, x_{27}) to those of (x_5, x_{26}) . Then

$$|\text{Hom}(Q(L_n), T)| = \text{tr}(A_T M_T^n) \quad (n \geq 1). \quad (2)$$

For $n = 0$ the right-hand side counts colorings of the clasp closed up by $x_5 = x_3$, $x_{26} = x_{27}$. Its relations become $x_1 \triangleright x_{26} = x_2$, $x_{26} \triangleright x_1 = x_{26}$, $x_2 \triangleright x_3 = x_1$, $x_4 \triangleright x_1 = x_3$ and $x_3 \triangleright x_3 = x_4$; the last one is a kink ($x_4 = x_3$), and the others are the relations of H under $x_1 \mapsto x_3$, $x_2 \mapsto x_2$, $x_{26} \mapsto x_1$, $x_3 \mapsto x_4$. So (2) holds for all $n \geq 0$ with $L_0 := H$. (We also verified $\text{tr } A_T = |\text{Hom}(Q(H), T)|$ and $R_T^\top = A_T M_T$ directly for $(\mathbb{Z}_3)^2$, $(\mathbb{Z}_5)^2$, $(\mathbb{Z}/4)^2$, $(\mathbb{F}_4)^2$, $(\mathbb{Z}/2)^4$ and T_8 .)

Lemma 5.4 (Corrected transfer lemma). *Suppose that for every $w \in T^2$ the block B has a coloring with input colors w and output colors w , i.e. $M_T[w, w] \geq 1$. Then $|\text{Hom}(Q(L_{n+1}), T)| \geq |\text{Hom}(Q(L_n), T)|$ for every $n \geq 0$ (with $L_0 = H$).*

Proof. Given a coloring of L_n , let w be the colors of the two arcs entering the clasp (for $n = 1$ these are the arcs x_3, x_{27} leaving c_9 and c_{44}). Insert one more copy of T_0 at this place, colored by a coloring with inputs and outputs w , and keep all other colors. This is a coloring of L_{n+1} , and different colorings of L_n give different colorings of L_{n+1} . $\square$

The hypothesis is a finite check. We verified it for $(\mathbb{Z}_2)^2$, $(\mathbb{Z}_3)^2$, $(\mathbb{Z}_5)^2$, $(\mathbb{Z}_7)^2$, $(\mathbb{Z}/4)^2$, $(\mathbb{F}_4)^2$, $(\mathbb{Z}/2)^4$ and T_8 (for $(\mathbb{Z}_p)^2$ it suffices to check one pair w in each orbit of $SL(2, p)$ on T^2 , by Lemma 4.5).

Corollary 5.5. *The counting invariants of $(\mathbb{Z}_7)^2$ and of $(\mathbb{F}_4)^2$ distinguish H from every Allen–Swenberg link L_n , $n \geq 1$. The counting invariant of $(\mathbb{Z}_5)^2$ distinguishes H from every L_n with $n \geq 2$.*

Proof. By Lemma 5.4,

$$|\mathrm{Hom}(Q(L_n), (\mathbb{Z}_7)^2)| \geq |\mathrm{Hom}(Q(L_1), (\mathbb{Z}_7)^2)| = 123025 > 4753 = |\mathrm{Hom}(Q(H), (\mathbb{Z}_7)^2)|,$$

$|\mathrm{Hom}(Q(L_n), (\mathbb{F}_4)^2)| \geq 3376 > 496$, and $|\mathrm{Hom}(Q(L_n), (\mathbb{Z}_5)^2)| \geq 13225 > 1225$ for $n \geq 2$. The lower bounds only require the explicit colorings found by the solver, each of which is checked against all relations. $\square$

Table 3 lists $|\mathrm{Hom}(Q(L_n), T)|$ for $n \leq 5$ computed from (1); we computed them up to $n = 8$. In every column the numbers are strictly increasing in n up to $n = 8$, so each of these quandles separates $L_1, \dots, L_8$ from each other (and $(\mathbb{Z}_5)^2$ separates $L_2, \dots, L_8$ from H and from L_1).

	$(\mathbb{Z}_5)^2$	$(\mathbb{Z}_7)^2$	$(\mathbb{Z}/4)^2$	$(\mathbb{Z}/2)^4$	$(\mathbb{F}_4)^2$	T_8
H	1 225	4 753	736	1 216	496	224
L_1	1 225	123 025	2 176	48 736	3 376	944
L_2	13 225	7 466 641	25 216	4 800 736	156 496	12 464
L_3	37 225	481 705 105	393 856	480 000 736	7 335 376	196 784
L_4	7 723 225	32 264 735 377	6 292 096	48 000 000 736	357 795 496	3 145 904
L_5	38 371 225	2 234 220 779 665	100 663 936	4 800 000 000 736	17 413 785 376	50 331 824

Table 3: $|\mathrm{Hom}(Q(L_n), T)|$ from the transfer formula (1). Over $(\mathbb{Z}_2)^2$ and $(\mathbb{Z}_3)^2$ all values equal those of H (28 and 153). The columns $(\mathbb{Z}/4)^2$ and $(\mathbb{Z}/2)^4$ are $640 + 96 \cdot 16^n$ and $736 + 480 \cdot 100^n$, and T_8 is $176 + 48 \cdot 16^n$ (Theorems 5.6, 5.7, 5.8).

5.7 Small symplectic quandles that separate the whole family

Theorem 5.6. *Let $T = (\mathbb{Z}/4)^2$ with the standard form. For every $n \geq 1$,*

$$|\mathrm{Hom}(Q(L_n), T)| = 640 + 96 \cdot 16^n, \quad |\mathrm{Hom}(Q(H), T)| = 736 = 640 + 96 \cdot 16^0,$$

and

$$\Phi_E(L_n, T) = 16q + 216q^2 + 192q^3 + 120q^4 + 48q^5 + 48q^6 + 96 \cdot 16^n q^{13},$$

where the same formula with $n = 0$ gives $\Phi_E(H, T)$. In particular the plain counting invariant of this 16-element quandle distinguishes $H, L_1, L_2, \dots$ pairwise.

Proof. The transfer matrix M_T is a 256×256 integer matrix. We computed M_T and R_T exactly and verified, in exact integer arithmetic, that

$$M_T^2 = 17M_T - 16I,$$

so the minimal polynomial of M_T divides $(x - 1)(x - 16)$ (it equals it, since $M_T \neq I, 16I$). By (2) the sequence $c_n = |\mathrm{Hom}(Q(L_n), T)|$ satisfies $c_{n+2} = 17c_{n+1} - 16c_n$ for all $n \geq 0$, and so does $640 + 96 \cdot 16^n$. Both sequences start with $c_0 = 736$ and $c_1 = 2176$ (the value for H also agrees with a brute-force count over all 16^4 assignments), hence they coincide.

For the enhanced polynomial we enumerated the 187 subquandles $S \subseteq T$ (87 trivial and 100 non-trivial). For a trivial S , $|\mathrm{Hom}(Q(L_n), S)| = |S|^3$. For each non-trivial S we computed the transfer matrices of colorings with values in S and their exact minimal polynomials, which all divide

$(x-1)(x-16)$. Writing $e_n(S) = \#\{f : \text{Im}(f) = S\}$, we have $|\text{Hom}(Q(L_n), S)| = \sum_{S' \subseteq S} e_n(S')$, and Möbius inversion over the lattice of subquandles gives $e_n(S)$ for all n as a combination of 1^n and 16^n . Summing over $|S| = m$ gives the coefficients above, which we checked for $n \leq 12$; since they satisfy the same recurrence, they hold for all n . Direct computations of Φ_E for H, L_1, L_2, L_3 agree. $\square$

The 13-element subquandle carrying all additional colorings is $\{0\} \cup \{\text{vectors with a unit coordinate}\}$. The values at $n = 0$ are those of H because of (2): H is literally the “zeroth” Allen–Swenberg link.

Theorem 5.7. *Let $T = (\mathbb{Z}/2)^4$ with the form $J \oplus J$. For every $n \geq 1$, $|\text{Hom}(Q(L_n), T)| = 736 + 480 \cdot 100^n$, while $|\text{Hom}(Q(H), T)| = 1216 = 736 + 480 \cdot 100^0$. Moreover*

$$\begin{aligned} \Phi_E(L_n, T) = & 16q + 360q^2 + 360q^3 + 480q^4 + 720(16^n - 1)q^7 + 360(49^n - 1)q^{10} \\ & + 360(49^n - 2 \cdot 16^n + 1)q^{11} + 360(100^n - 16^n - 49^n + 1)q^{15} \\ & + 120(100^n + 3 \cdot 16^n - 3 \cdot 49^n - 1)q^{16}, \end{aligned}$$

and for $n = 0$ this is $\Phi_E(H, T) = 16q + 360q^2 + 360q^3 + 480q^4$.

Proof. As for Theorem 5.6: the transfer matrix satisfies $M^2 = 101M - 100I$, and the subquandle lattice (405 subquandles) gives recurrences dividing $(x-1)(x-16)(x-49)(x-100)$. The value at $n = 0$ is that of H by (2). $\square$

The smallest example we found is degenerate.

Theorem 5.8. *Let $T_8 = (\mathbb{Z}/2)^3$ with the degenerate form $J \oplus 0$, i.e. $\langle x, y \rangle = x_1y_2 + x_2y_1$. For every $n \geq 0$ (with $L_0 = H$),*

$$|\text{Hom}(Q(L_n), T_8)| = 176 + 48 \cdot 16^n, \quad |\text{Hom}(Q(H), T_8)| = 224 = 176 + 48 \cdot 16^0,$$

and

$$\Phi_E(L_n, T_8) = 8q + 96q^2 + 72q^3 + 48q^4 + 48(16^n - 1)q^7,$$

Hence already this 8-element quandle distinguishes $H, L_1, L_2, \dots$ pairwise.

Proof. The transfer matrix is 64×64 and satisfies $M^2 = 17M - 16I$; T_8 has 59 subquandles and all restricted transfer matrices have minimal polynomials dividing $(x-1)(x-16)$. The argument is then the same as for Theorem 5.6; direct computations of Φ_E for H, L_1, L_2, L_3 agree. $\square$

Writing elements of T_8 as (v, t) with $v \in \mathbb{F}_2^2$, $t \in \mathbb{F}_2$, the operation is $(v, t) \triangleright (w, s) = (v \triangleright w, t + \langle v, w \rangle s)$. So T_8 is an extension of the 4-element quandle $(\mathbb{F}_2)^2$, which by itself sees nothing (Table 3), by a module of the kind used for the cocycle invariants of [5]. The detection thus comes from a “cocycle” layer on top of a quandle that is blind to the Allen–Swenberg block.

For $(\mathbb{F}_4)^2$ the transfer matrix has minimal polynomial $(x-1)(x-6)(x-11)(x+9)(x-41)(x-49)(x^4 + x^3 + 16x^2 + 86x + 521)$; the counts are listed in Table 3. Here, unlike in all cases we computed over $(\mathbb{Z}_p)^2$, additional colorings also occur with $\mu \mapsto v \neq 0$ (for L_2).

5.8 Other symplectic quandles

Table 4 summarizes the other symplectic quandles we tried on H, L_1, L_2 . The $(\mathbb{Z}_2)^2$, $(\mathbb{Z}_3)^2$ and $(\mathbb{Z}/6)^2$ quandles do not separate H, L_1, L_2 : for $(\mathbb{Z}_2)^2$ and $(\mathbb{Z}_3)^2$ the block B is “transparent” (its transfer matrix is the identity), and $(\mathbb{Z}/6)^2 \cong (\mathbb{Z}_2)^2 \times (\mathbb{Z}_3)^2$. Degenerate forms behave differently from case to case: λJ with λ a zero divisor of $\mathbb{Z}/4$ or $\mathbb{Z}/6$ and $J \oplus 0$ over $\mathbb{Z}/3$ do not separate H, L_1, L_2 , whereas $J \oplus 0$ over $\mathbb{Z}/2$ does (Theorem 5.8).

T	$ T $	$ \text{Hom}(Q(H), T) $	$ \text{Hom}(Q(L_1), T) $	$ \text{Hom}(Q(L_2), T) $	separates
$(\mathbb{Z}/4)^2, J$	16	736	2 176	25 216	yes
$(\mathbb{Z}/4)^2, 2J$ (degenerate)	16	1 792	1 792	1 792	no
$(\mathbb{Z}/6)^2, J$	36	4 284	4 284	4 284	no
$(\mathbb{Z}/6)^2, 2J$ (degenerate)	36	9 792	9 792	9 792	no
$(\mathbb{Z}/6)^2, 3J$ (degenerate)	36	20 412	20 412	20 412	no
$(\mathbb{F}_4)^2, J$	16	496	3 376	156 496	yes
$(\mathbb{Z}/2)^4, J \oplus J$	16	1 216	48 736	4 800 736	yes
$(\mathbb{Z}/2)^3, J \oplus 0$ (degenerate)	8	224	944	12 464	yes
$(\mathbb{Z}/3)^3, J \oplus 0$ (degenerate)	27	4 131	4 131	4 131	no

Table 4: Counting invariants for further symplectic quandles. “Separates” means that H, L_1, L_2 have pairwise different counts.

5.9 Consequences and outlook

The corrected picture is as follows. The enhanced polynomial over $(\mathbb{Z}_3)^2$ and $(\mathbb{Z}_5)^2$ does not separate L_1 from H , so the main claim of version 1 was an artifact of counting arc colors. Nevertheless symplectic quandles do detect causality on the Allen–Swenberg testbed: the 16-element quandles $(\mathbb{Z}/4)^2$ and $(\mathbb{Z}/2)^4$ and the 8-element quandle T_8 separate H and all L_n pairwise by the plain counting invariant, with closed formulas valid for all n ; $(\mathbb{Z}_7)^2$ and $(\mathbb{F}_4)^2$ separate H from all L_n , and $(\mathbb{F}_4)^2$ also separates $L_1, \dots, L_8$ from each other. Over $(\mathbb{Z}_p)^2$ the corrected enhancement adds at most one number to the count, because by Lemma 4.7 the image of a coloring is either a set of collinear vectors or all of $T \setminus \{0\}$ or T ; in our computations it adds nothing. Over rings such as $\mathbb{Z}/4$ the lattice of subquandles is richer, and the enhancement does record additional structure (the 13-element subquandle in Theorem 5.6).

Conjecture 5.9. *For every $n \geq 1$ the set of primes p for which $(\mathbb{Z}_p)^2$ distinguishes L_n from H is infinite, and in fact has positive density. For $n = 1$ it contains 7, 11, 17, 19, 23, 31 and does not contain 2, 3, 5, 13; for $n = 2$ it contains 5, 7, 11 and does not contain 2, 3.*

A natural approach is to identify the colorings with $\mu \mapsto 0$ with parabolic representations of the group of the sky sublink $K_1 \cup K_2$ into $SL(2, \mathbb{F}_p)$. If the additional colorings come from a part of this representation variety that is defined over a number field K but not over $\mathbb{Q}$ (for instance a finite set of points, as for the parabolic representations of two-bridge links), then they exist over $\mathbb{F}_p$ exactly for the primes with a suitable splitting behaviour in K . By Chebotarev’s density theorem this is a set of positive density, which need not contain all large primes; this would explain exceptional primes such as 13 and 29. We have not tried to determine this variety.

Reproducibility. All code and data for this version are openly available at <https://github.com/amirbekbakhshiloev/symplectic-quandle-amirbek-updated>. The Python package `symq` contains: the presentations of H, L_1, L_2 and the generator of L_n ; the symplectic quandles over $\mathbb{Z}/n$ and $\mathbb{F}_{p^k}$ with axiom checks; the coloring solver (Python, with a C accelerator implementing the same algorithm); the corrected and the arc-color polynomials; the Reidemeister II example; the topological checks (components, linking numbers, Alexander polynomials, planar reconstruction and Kauffman bracket); the transfer matrices and the exact minimal-polynomial certificates. The script `reproduce.py` regenerates every number and table in this paper; its output is stored in the directory `results`.

6 Conclusion

This paper examined whether symplectic quandle colorings can detect causal structure encoded by sky links, using the connected sum of two Hopf links H and the Allen–Swenberg links L_n as a testbed. The first version used the number of distinct colors on the arcs of a diagram in place of the size of the image of a coloring. This quantity is not invariant under Reidemeister moves, and with the correct definition the separation of H from L_1 over $(\mathbb{Z}_3)^2$ and $(\mathbb{Z}_5)^2$ disappears: the extra arc colors of the Allen–Swenberg diagrams all lie in the subquandle generated by very few colors.

After correcting the definition and verifying the link data (Alexander polynomials, a unique planar realization of the Wirtinger presentation of L_1 , and Kauffman-bracket spans 116, 216, 316 for L_1, L_2, L_3), we obtained the following.

- Over $(\mathbb{Z}_p)^2$ the enhanced polynomial adds at most one number to the counting invariant, and nothing in our computations; the count separates L_1 from H for $p = 7, 11, 17, 19, 23, 31$ but not for $p = 2, 3, 5, 13$.
- A corrected transfer lemma shows that counts are non-decreasing along the family whenever the Allen–Swenberg block admits identity colorings; hence $(\mathbb{Z}_7)^2$ separates H from every L_n .
- For the 16-element symplectic quandle $(\mathbb{Z}/4)^2$ we proved $|\text{Hom}(Q(L_n), T)| = 640 + 96 \cdot 16^n$ for all $n \geq 1$ and 736 for H , and computed the corrected enhanced polynomial of every L_n ; for $(\mathbb{Z}/2)^4$ the count is $736 + 480 \cdot 100^n$, and for the 8-element degenerate quandle T_8 it is $176 + 48 \cdot 16^n$ versus 224.

In particular symplectic quandles, used correctly, do detect causality on the Allen–Swenberg testbed and distinguish the Allen–Swenberg links from each other, and a very small target quandle suffices.

Limitations. Our statements concern the specific diagrams of Allen and Swenberg, whose relation to actual skies of causally related events is itself conjectural. Quandle counting is computationally expensive for large targets; for the family L_n this is overcome by the transfer matrix, but for general sky links the transfer structure is not available. Finally, Conjecture 5.9 is supported only by computations for primes up to 31.

Acknowledgements

Conducted in Summer 2025 as part of the Horizon Academic Research Program (Advanced Mathematics), and revised in 2026. This project was supervised by Professor Vladimir Chernov of Dartmouth College and Dr. Ryan Maguire of MIT. I thank them both for their continual support, and I thank Zining Fan and Vladimir Chernov for finding and explaining the error in the first version.

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A Wirtinger presentations

Each entry $k: (a, b, c)$ stands for the relation $x_a \triangleright x_b = x_c$ at crossing c_k . For L_2 , the crossings $c_1, \dots, c_{45}$ agree with those of L_1 except $c_9: (6, 28, 47)$ and $c_{44}: (45, 40, 46)$; the block B consists of $c_{46}, \dots, c_{85}$.

L_1 (crossings $c_1, \dots, c_{45}$).

1: (1, 26, 2)	2: (27, 1, 26)	3: (2, 3, 1)	4: (4, 1, 3)	5: (5, 3, 4)
6: (6, 7, 5)	7: (7, 6, 8)	8: (28, 6, 29)	9: (6, 28, 3)	10: (8, 7, 9)
11: (10, 9, 7)	12: (9, 10, 11)	13: (12, 11, 10)	14: (11, 12, 13)	15: (14, 13, 12)
16: (15, 14, 19)	17: (15, 13, 22)	18: (16, 14, 20)	19: (16, 13, 21)	20: (17, 21, 14)
21: (18, 21, 13)	22: (17, 22, 20)	23: (18, 22, 19)	24: (23, 20, 21)	25: (24, 20, 22)
26: (23, 19, 26)	27: (24, 19, 25)	28: (28, 29, 30)	29: (31, 30, 29)	30: (30, 31, 32)
31: (33, 32, 31)	32: (32, 33, 34)	33: (35, 34, 33)	34: (36, 35, 40)	35: (36, 34, 38)
36: (37, 35, 41)	37: (37, 34, 39)	38: (42, 39, 35)	39: (43, 39, 34)	40: (42, 38, 41)
41: (43, 38, 40)	42: (44, 40, 25)	43: (44, 41, 39)	44: (45, 40, 27)	45: (45, 41, 38)

Block B of L_2 (crossings $c_{46}, \dots, c_{85}$).

46: (67, 48, 47)	47: (48, 67, 49)	48: (68, 67, 69)	49: (67, 68, 3)	50: (49, 48, 50)
51: (51, 50, 48)	52: (50, 51, 52)	53: (53, 52, 51)	54: (52, 53, 54)	55: (55, 54, 53)
56: (56, 55, 58)	57: (56, 54, 61)	58: (57, 55, 59)	59: (57, 54, 60)	60: (62, 60, 55)
61: (63, 60, 54)	62: (62, 61, 59)	63: (63, 61, 58)	64: (64, 58, 46)	65: (64, 59, 60)
66: (65, 58, 66)	67: (65, 59, 61)	68: (68, 69, 70)	69: (71, 70, 69)	70: (70, 71, 72)
71: (73, 72, 71)	72: (72, 73, 74)	73: (75, 74, 73)	74: (76, 75, 78)	75: (76, 74, 81)
76: (77, 75, 79)	77: (77, 74, 80)	78: (82, 80, 75)	79: (83, 80, 74)	80: (82, 81, 79)
81: (83, 81, 78)	82: (84, 78, 66)	83: (84, 79, 80)	84: (85, 78, 27)	85: (85, 79, 81)